

P100-082: Soil respiration dataset for European forest sites evaluating holistic management practices

Principia-100 | Standardized portfolio | 1 October 2026

Scenario and evaluation contract

Domain: Forest ecology. The current default task predicts Author-derived forest-soil CO2 mass flux on reserved future dates in g CO2 m⁻² h⁻¹. Independent unit: complete reserved future temporal block at each calibrated site. Its packaged cohort contains 6,928 assigned rows in 15 groups.

All supplied outcomes are now exposed. Historical reserved evaluations remain part of the evidence record; future scores are retrospective. No newly established fundamental law or measured deployment impact is admitted by this packaging.

Current portfolio findings

Measurement audit: Dimensionless parameter profiling repairs the thermal optimizer but does not create new ecological evidence.

Limit: Corrected E 0≈208.04 K and near-zero profile gradient repair the initial near 200 K fit; activation-mixture/history terms add negligible accuracy. Do not attribute optimizer repair to a new biological mechanism.

Retrospective extension: Directional past wetting adds a small forward-prediction improvement compatible with known rewetting effects.

Limit: 28/48 blocks and 12/16 label means improve. Reoptimizing a no-pulse control also improves; not all 1.60 %gain is wetting. Season, substrate and sampling interval remain confounded.

Versioned tasks

Task family	Target and information	Source scope
original	Author-derived forest-soil CO2 mass flux on reserved future dates (g CO2 m ⁻² h ⁻¹)	original corpus
round2	Author-derived forest-soil CO2 mass flux on reserved future dates (g CO2 m ⁻² h ⁻¹)	original corpus

Different targets, calibration budgets and cohorts have different task IDs. Their raw error values cannot be pooled into a scenario ranking. Exact equations, coefficients, permitted variables, timing, groups and per-model results are bound in the task packages.

Evaluation: the shared CLI provides numerical scoring, explicit coverage and abstention, uncertainty and event diagnostics. Agent review separately assesses scientific support; no aggregate discovery score is produced.

Detailed reference note: Predicting forest-soil respiration at calibrated sites

Principia-100 · P100-082 · Frozen scientific reference, 30 September 2026

1. Scenario and available measurements

The HoliSoils release contains 50,140 chamber observations, environmental measurements and management contexts from 2021-2025. Its 19 site labels mostly represent European forests; RincondelBatovi is in Uruguay. Flux is an author-calibrated chamber-slope product in $\text{g CO}_2 \text{ m}^{-2} \text{ h}^{-1}$. Native metadata also document soil temperature, volumetric water content, management and root-exclusion trenching. Raw chamber concentration time series are not supplied.

This task predicts a future chamber observation at an already calibrated site, using same-date environmental conditions and known context. It addresses interpretable environmental response models for monitoring. It does not evaluate unseen-site transfer, annual carbon budgets or causal management effects.

2. Experimental method

Inputs are native 5 cm soil temperature, volumetric moisture, day of year, site and subsite/trenching context. Rows require finite temperature within -15 to 50 degrees Celsius, moisture missing or within 0-100 native percent, and finite flux. All negative and unusually high finite flux values remain scored. The input-defined scope retains 38,389 rows; three Dutch sites lack 5 cm temperature entirely.

The latest 20% of native dates per site were reserved before fitting. Development contains 31,461 eligible rows at 16 sites; confirmation contains 6,928 rows at 15 sites. Dobroc has no eligible temperature input on its reserved dates, so it contributes no final score and is not resplit. Three forward development folds fit only earlier dates; all points and contexts on a date remain together. Context calibration, missingness treatment and flexible-model tuning use training data only.

Five cycles test moisture optimum, dry saturation, thermal-moisture interaction, site heterogeneity and seasonal response. The frozen selection favors a simpler domain model when its development error is within 1% of the best extension. Confirmation is scored in a separate process after selection; these targets are now exposed. Response slope, intercept, slope confidence interval and chamber-conversion quantities are forbidden predictors.

3. Reference equation and physical interpretation

The selected reference is the established Lloyd-Taylor temperature response. For a calibrated context c , with soil temperature T in degrees Celsius:

$$\widehat{R}_c(T) = A_c \exp \left[E_0 \left(\frac{1}{10 + 46.02} - \frac{1}{T + 46.02} \right) \right]$$

The reference coefficient is $E_0 = 200.000005 \text{ K}$. Context amplitudes A have the same flux units as R ; all 96 fitted amplitudes and site/global fallbacks are disclosed in `rules.json`. The denominator offsets represent temperature differences in kelvin, so the exponent is dimensionless. A is the fitted response at 10 degrees Celsius.

This is an apparent ecological temperature response. It combines microbial, root, substrate and transport effects rather than identifying one biochemical activation energy. A computational audit found that thermal-parameter scaling limited optimization and left E_0 near its starting value. The unchanged frozen equation is therefore a near-200 K calibrated reference, not a claim of globally optimized thermal coefficients. A development-only profile found a small objective improvement near 210 K; no final data guided a revision.

4. Findings and predictive performance

The primary metric is mean RMSE over the 15 whole reserved site blocks. Each site receives equal weight regardless of its row count; multiple chambers and dates do not become independent sites.

Frozen model	Mean site RMSE (g CO ₂ m ⁻² h ⁻¹)
Selected Lloyd-Taylor reference	0.287474
Thermal-moisture interaction	0.287676
Q10 temperature response	0.291073
Calibrated context mean	0.333953
Fixed kernel comparator	0.292475
Nested-tuned kernel comparator	0.290985

The reference improves mean error by 13.92% over context means, winning 14 of 15 sites. Its gain over Q10 is 1.24%; gains over kernel comparators are also below 2%. It fails the 5% improvement gate against every baseline. Site RMSE spans approximately 0.119-0.577 in native flux units, which exposes meaningful differences in reliability across systems.

The moisture interaction was the best ASD extension in development, but its gain over the simpler temperature reference was below 1%; on reserved sites their errors differ by only 0.000202. Added site-specific slopes and seasonal coefficients did not improve development evidence. These negative results constrain claims that extra mechanisms automatically yield better transfer. The frozen simpler selection is retained even though each comparison has different group-level strengths.

5. Practical value and limits

The reference provides a transparent, executable monitoring response with declared site calibration. Its value is a reproducible performance standard and evidence for the limits of added environmental complexity. It cannot replace sensor calibration or independent ecological measurements. A fitted context offset does not establish treatment causality or isolate root respiration.

Instrument differences, recurring collars, site/date correlations, missing moisture and unverified cross-site moisture conventions limit interpretation. No percent/fraction correction was invented. Negative flux and author-derived values are visible in the evidence. Coverage excludes rows without valid 5 cm inputs and all final observations at Dobroc. Fifteen calibrated systems do not establish universal geographic transfer; uncertainty cannot be estimated by treating 6,928 rows as independent replicates.

6. Evidence and use

`run.py` reproduces all frozen predictions and grouped errors; `rules.json` exposes every coefficient and calibration term. `data/` separates allowed predictors from source-row-anchored responses. `evidence/` and the evaluator support comparisons with other declared ASD outputs. `research_history/` retains all five cycles, date/eligibility manifests, prior art, freeze and optimizer qualification. Further score-guided revisions require fresh confirmation evidence.

Source: Lehtonen, Mäkipää and colleagues, [HoliSoils v1](#), DOI 10.5281/zenodo.17519671. Prior art: [Lloyd and Taylor \(1994\)](#) and [Ruehr et al. \(2010\)](#). Classification: established-model reproduction, reference comparison and preserved negative extension evidence; no adjudicated novelty.